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% =====
% Combined TE and TM Propagation Loss Analysis using Cut-Back Method
%
% This script:
% 1. Loads TE and TM datasets
% 2. Performs polynomial fitting
% 3. Performs cut-back linear regression
% 4. Calculates propagation loss
% 5. Generates grouped comparison plots:
%
%     Figure 1(a): TE Optical Spectra
%     Figure 1(b): TM Optical Spectra
%
%     Figure 2(a): TE Cut-back Linear Regression
%     Figure 2(b): TM Cut-back Linear Regression
%
%     Figure 3: TE vs TM Loss Comparison
%
% =====

clear;
clc;
close all;

dbstop if error

% =====
% SETTINGS
% =====
FONTSIZE = 13;
LEGENDSIZE = 8;

lambda0 = 1.55e-6;
PORT = 2;

Error_Intervals = exist('regress','file');

```

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% =====
% COMMON DEVICE LENGTHS
% =====
Num = [0, 0.5, 1.0, 3.0];

% =====
% TE FILES
% =====
files_TE = { ...
    'LukasC_SpiralWG0kTE_1297.mat', ...
    'LukasC_SpiralWG5kTE_1300.mat', ...
    'LukasC_SpiralWG10kTE_1298.mat', ...
    'LukasC_SpiralWG30kTE_1299.mat' ...
};

url_TE = { ...
    'https://www.dropbox.com/s/1v5wzon5nexggn6/LukasC_SpiralWG0kTE_1297.mat?dl=1', ...
    'https://www.dropbox.com/s/7r8svd1ukjae8ox/LukasC_SpiralWG5kTE_1300.mat?dl=1', ...
    'https://www.dropbox.com/s/kfsv1yeghue0i38/LukasC_SpiralWG10kTE_1298.mat?dl=1', ...
    'https://www.dropbox.com/s/5o66l3xd7rnxtgq/LukasC_SpiralWG30kTE_1299.mat?dl=1' ...
};

% =====
% TM FILES
% =====
files_TM = { ...
    'LukasC_SpiralWG0kTM_1293.mat', ...
    'LukasC_SpiralWG5kTM_1296.mat', ...
    'LukasC_SpiralWG10kTM_1294.mat', ...
    'LukasC_SpiralWG30kTM_1295.mat' ...
};

url_TM = { ...
    'https://www.dropbox.com/s/anlo8zmydrji1f8/LukasC_SpiralWG0kTM_1293.mat?dl=1', ...
    'https://www.dropbox.com/s/yvdfgl8qoq3d0fz/LukasC_SpiralWG5kTM_1296.mat?dl=1', ...
    'https://www.dropbox.com/s/rossgslht5r5cq7/LukasC_SpiralWG10kTM_1294.mat?dl=1', ...
    'https://www.dropbox.com/s/vj4uf6u59h9vt60/LukasC_SpiralWG30kTM_1295.mat?dl=1' ...
};

```

```

    };

% =====
% DOWNLOAD FILES IF NEEDED
% =====
disp('Checking TE files...')

```

Checking TE files...

```

for i = 1:length(files_TE)

    if ~exist(files_TE{i}, 'file')

        fprintf('Downloading %s\n', files_TE{i});

        websave(files_TE{i}, url_TE{i});

    end

end

disp('Checking TM files...')

```

Checking TM files...

```

for i = 1:length(files_TM)

    if ~exist(files_TM{i}, 'file')

        fprintf('Downloading %s\n', files_TM{i});

        websave(files_TM{i}, url_TM{i});

    end

end

```

```

% =====
% FUNCTION HANDLE FOR PROCESSING
% =====
processMode = @(files) process_dataset(files, PORT);

% =====
% PROCESS TE DATA
% =====
[lambda_TE, amplitude_TE, amplitude_poly_TE] = processMode(files_TE);

% =====
% PROCESS TM DATA
% =====
[lambda_TM, amplitude_TM, amplitude_poly_TM] = processMode(files_TM);

% =====
% FIGURE 1: OPTICAL SPECTRA
% =====
figure('Position',[100 100 1400 700]);

% -----
% FIGURE 1(a) TE
% -----
subplot(1,2,1)

LegendText = {};

for i = 1:length(files_TE)

    plot(lambda_TE*1e6, amplitude_TE(:,i));
    hold on;

    plot(lambda_TE*1e6, amplitude_poly_TE(:,i), 'LineWidth',2);

    LegendText{2*i-1} = ['Raw ' num2str(Num(i)) ' cm'];
    LegendText{2*i}   = ['Fit ' num2str(Num(i)) ' cm'];

```

```

end

title('(a) TE Mode Optical Spectra')

xlabel('Wavelength (\mum)')

ylabel('Optical Power (dBm)')

legend(legendText, ...
    'Location','eastoutside', ...
    'FontSize',LEGENDSIZE)

grid on
axis tight
set(gca,'FontSize',FONTSIZE)

% -----
% FIGURE 1(b) TM
% -----

subplot(1,2,2)

legendText = {};

for i = 1:length(files_TM)

    plot(lambda_TM*1e6, amplitude_TM(:,i));
    hold on;

    plot(lambda_TM*1e6, amplitude_poly_TM(:,i), 'LineWidth',2);

    legendText{2*i-1} = ['Raw ' num2str(Num(i)) ' cm'];
    legendText{2*i}   = ['Fit ' num2str(Num(i)) ' cm'];

end

title('(b) TM Mode Optical Spectra')

```

```

xlabel('Wavelength (\mum)')

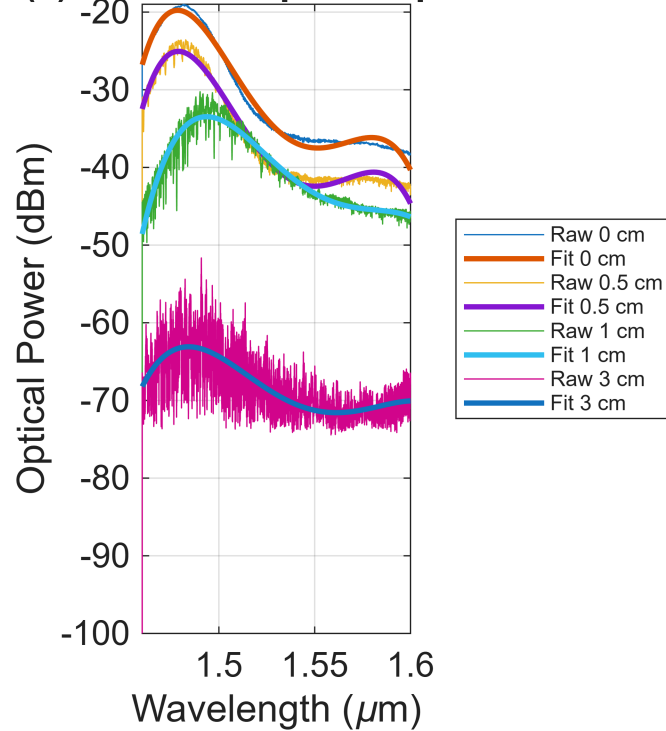
ylabel('Optical Power (dBm)')

legend(legendtext, ...
       'Location','eastoutside', ...
       'FontSize',LEGENDSIZE)

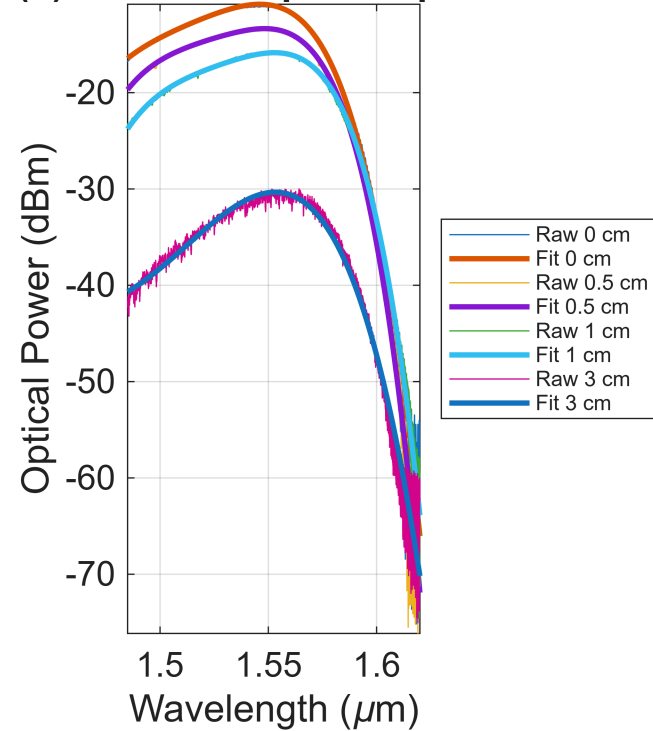
grid on
axis tight
set(gca,'FontSize',FONTSIZE)

```

(a) TE Mode Optical Spectra



(b) TM Mode Optical Spectra



% =====

```

% LINEAR REGRESSION
% =====
[~, index_TE] = min(abs(lambda_TE - lambda0));
[~, index_TM] = min(abs(lambda_TM - lambda0));

A_TE = [ones(length(Num),1) Num'] \ amplitude_poly_TE(index_TE,:);
A_TM = [ones(length(Num),1) Num'] \ amplitude_poly_TM(index_TM,:);

% =====
% FIGURE 2: CUT-BACK LINEAR REGRESSION
% =====
figure('Position',[120 120 1300 600]);

% -----
% FIGURE 2(a): TE
% -----
subplot(1,2,1)

plot(Num, amplitude_TE(index_TE,:), 'x', 'MarkerSize',10)
hold on

plot(Num, amplitude_poly_TE(index_TE,:), 'o', 'MarkerSize',8)

plot(Num, A_TE(1)+Num*A_TE(2), 'LineWidth',3)

title('(a) TE Cut-Back Regression')

xlabel('Waveguide Length (cm)')

ylabel('Loss (dB/cm)')

legend( ...
    'Raw Data', ...
    'Polyfit', ...
    'Linear Regression', ...
    'Location','best', ...
    'FontSize',LEGENDSIZE)

```

```

grid on
set(gca, 'FontSize', FONTSIZE)

% -----
% FIGURE 2(b): TM
% -----
subplot(1,2,2)

plot(Num, amplitude_TM(index_TM,:), 'x', 'MarkerSize', 10)
hold on

plot(Num, amplitude_poly_TM(index_TM,:), 'o', 'MarkerSize', 8)

plot(Num, A_TM(1)+Num*A_TM(2), 'LineWidth', 3)

title('(b) TM Cut-Back Regression')

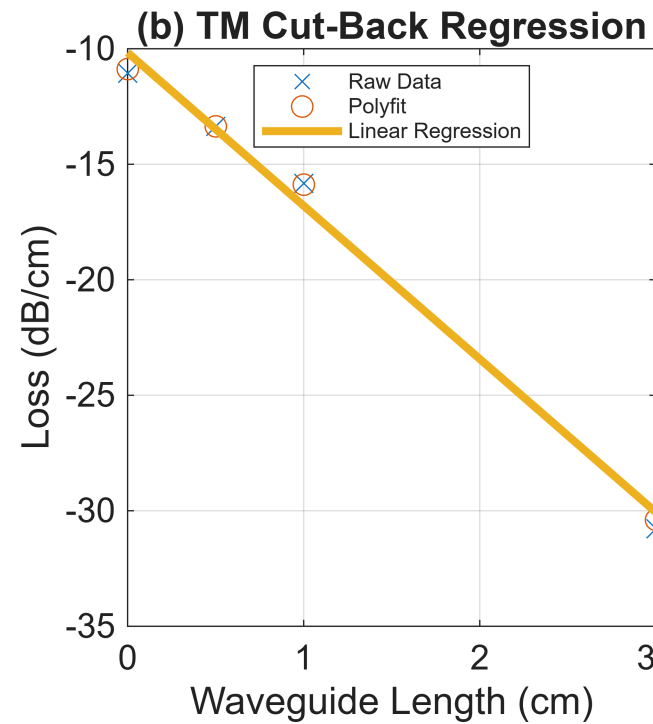
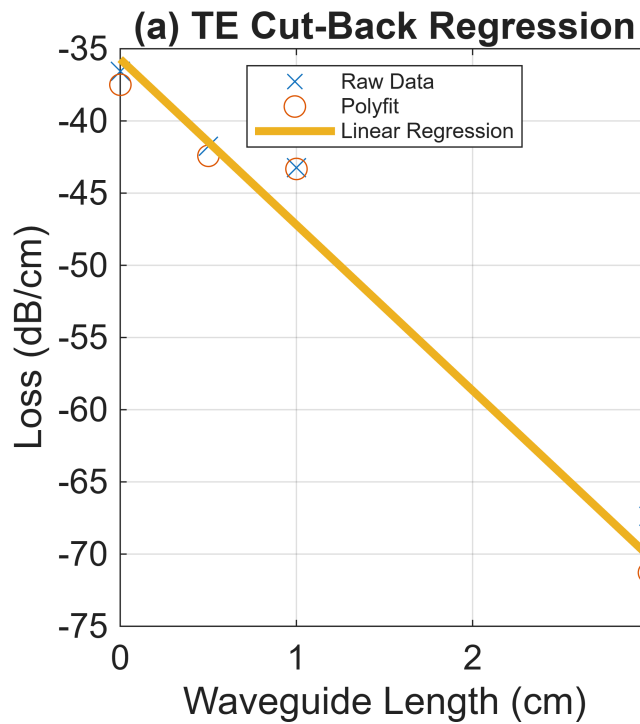
xlabel('Waveguide Length (cm)')

ylabel('Loss (dB/cm)')

legend( ...
    'Raw Data', ...
    'Polyfit', ...
    'Linear Regression', ...
    'Location', 'best', ...
    'FontSize', LEGENDSIZE)

grid on
set(gca, 'FontSize', FONTSIZE)

```

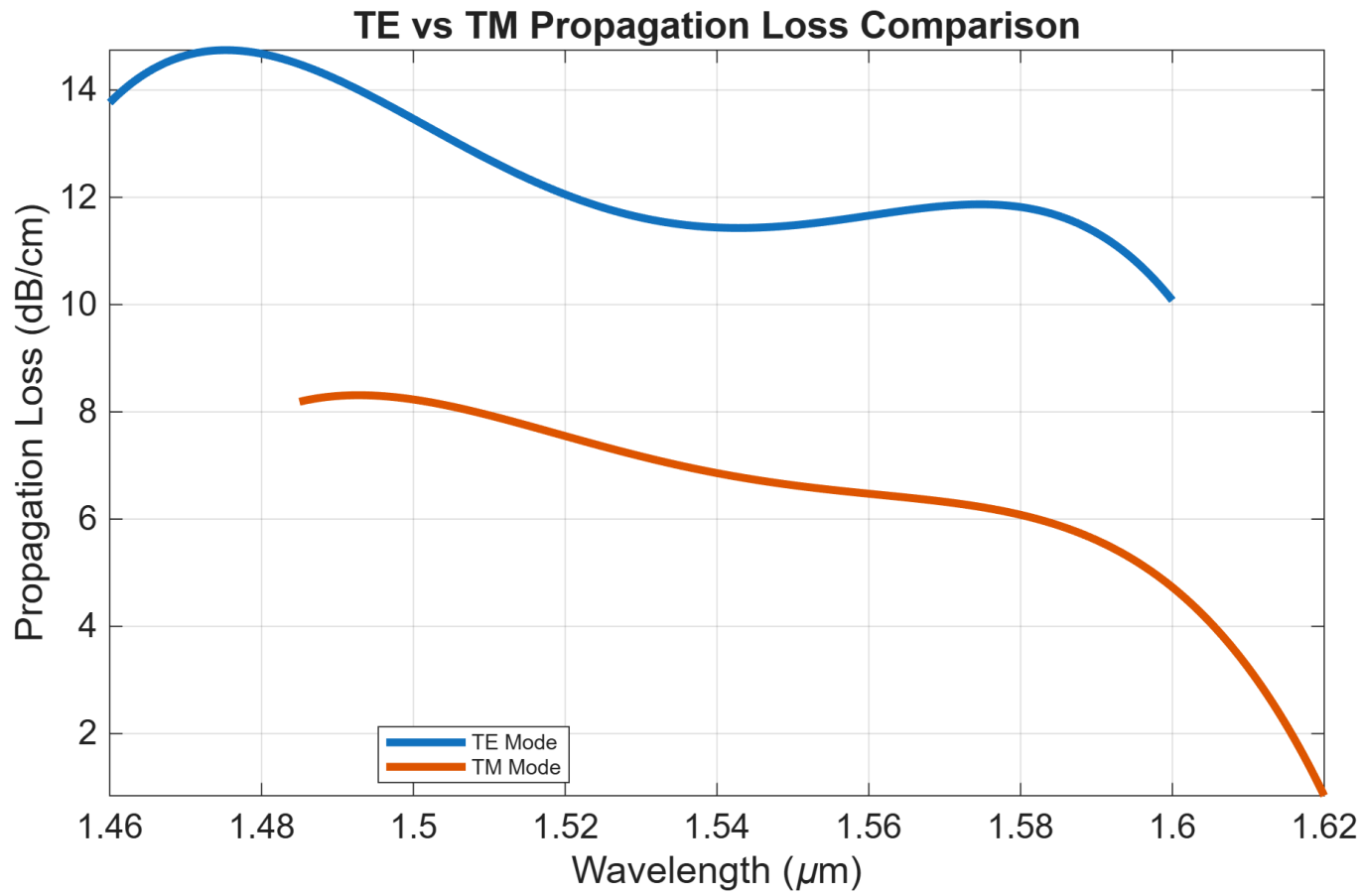
```
% =====
% WAVELENGTH DEPENDENT LOSS
% =====
D_TE = [ones(length(Num),1) Num'] \ amplitude_poly_TE';
D_TM = [ones(length(Num),1) Num'] \ amplitude_poly_TM';

% =====
% FIGURE 3: TE vs TM LOSS COMPARISON
% =====
figure('Position',[150 150 1200 700]);

plot(lambda_TE*1e6, -D_TE(2,:), 'LineWidth',3)
hold on

plot(lambda_TM*1e6, -D_TM(2,:), 'LineWidth',3)
```

```
xlabel('Wavelength (\mum)')  
ylabel('Propagation Loss (dB/cm)')  
title('TE vs TM Propagation Loss Comparison')  
legend( ...  
    'TE Mode', ...  
    'TM Mode', ...  
    'Location', 'best', ...  
    'FontSize', LEGENDSIZE)  
  
grid on  
axis tight  
  
set(gca, 'FontSize', FONTSIZE)
```



```
% =====
% DISPLAY RESULTS
% =====
disp('=====')
```

```
=====
```

```
disp('Propagation Loss Results at lambda0')
```

```
Propagation Loss Results at lambda0
```

```
disp('=====')
```

```
=====
```

```
disp(['TE Mode Loss = ' num2str(-A_TE(2), '%0.3f') ' dB/cm'])
```

```
TE Mode Loss = 11.477 dB/cm
```

```
disp(['TM Mode Loss = ' num2str(-A_TM(2), '%0.3f') ' dB/cm'])
```

```
TM Mode Loss = 6.632 dB/cm
```

```
disp('=====')
```

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% =====
```

```
% LOCAL FUNCTION
```

```
% =====
```

```
function [lambda, amplitude, amplitude_poly] = process_dataset(files, PORT)
```

```
    amplitude = [];
```

```
    amplitude_poly = [];
```

```
    for i = 1:length(files)
```

```
        load(files{i});
```

```
        lambda = scandata.wavelength(:);
```

```
        temp_power = scandata.power(:,PORT);
```

```
        temp_power = temp_power(:);
```

```
        amplitude(:,i) = temp_power;
```

```
xfit = (lambda - mean(lambda))*1e6;  
p = polyfit(xfit, amplitude(:,i), 4);  
amplitude_poly(:,i) = polyval(p, xfit);  
end  
end
```